

Complex Numbers (Algebra)

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Educational

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<https://sourceforge.net/u/rajatkalia/profile>
<https://www.youtube.com/@rajatkaliaeducation>

Store

<https://www.lulu.com/spotlight/rajatkalia11>
<https://amazon.com/author/rajatkalia>

Sports

<https://www.youtube.com/@indiangamersorganization>
<https://www.facebook.com/groups/igohn>
<https://india.voobly.com/>

Social

<https://www.goodreads.com/sardarbhagatsingh>
<https://x.com/RajatKalia1>
<https://www.linkedin.com/in/rajatkalia007/>
<https://www.facebook.com/rajatkalia007>
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<https://www.researchgate.net/profile/Rajat-Kalia>
<https://www.w3profile.com/rajatkalia/>
<https://www.reddit.com/user/rajatkalia/>

Blogs

<https://rk-e.blogspot.in/> [Physics]
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Freelancing

<https://www.guru.com/freelancers/rajat-kalia>
<https://www.codecademy.com/profiles/rajatkalia>
<https://leetcode.com/u/rajatkalia1/>
<https://www.scribd.com/user/308680256/Rajat-Kalia>

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Part I

Theory

Part II

Problem Set

Chapter 1

JEE Mains Problems

Q1: Let x and y be real numbers such that $50 \left(\frac{2x}{1+3i} - \frac{y}{1-2i} \right) = 31 + 17i$, $i = \sqrt{-1}$. Then the value of $10(x - 3y)$ is :

[JEE Mains 2nd Apr 2026 Shift 1]

(A) 20

(B) 31

(C) 35

(D) 75

{ Solution: <https://www.youtube.com/watch?v=7bvmAsRoeiU> }

Chapter 2

JEE Advanced Problems

Q1: Let $\mathbb{R}$ denote the set of all real numbers. Let $z_1=1+2i$ and $z_2=3i$ be two complex numbers, where $i=\sqrt{-1}$. Let

$$S=\{(x,y)\in\mathbb{R}\times\mathbb{R}|x+iy-z_1|=2|x+iy-z_2|\}.$$

Then which of the following statements is (are) TRUE?

[JEE Advanced 2025 Paper 1]

(A) S is a circle with centre $(-\frac{1}{3}, \frac{10}{3})$

(B) S is a circle with centre $(\frac{1}{3}, \frac{8}{3})$

(C) S is a circle with radius $\frac{\sqrt{2}}{3}$

(D) S is a circle with radius $\frac{2\sqrt{2}}{3}$

{ Solution : <https://www.youtube.com/watch?v=cYJl2HBn030> }